
Revealed Preferences of a Language Model: Experiment Design and Methods

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July 2026

Phil, this memo lays out the experiment design behind a revealed-preference estimation exercise I am running on a frontier language model, and it asks for your critique of that design before I scale it. I have written it as a methods memo to a knowledgeable peer, so I assume the standard behavioral apparatus (CRRA utility, prospect theory, reference dependence, loss aversion, MLE, likelihood-ratio testing) and spend my words instead on the parts that are specific to running an economics experiment on a language model, where the measurement problems are unfamiliar and, I think, where the design either succeeds or fails. Sections 1 through 7 describe the completed Phase 1 and Phase 2 instruments and estimates; Section 8 lays out the Phase 3 attribution design that is the point of the whole exercise; Section 9 is a short list of questions I would most value your view on.

1. Overview and research questions

The project performs revealed-preference structural estimation on a frontier language model treated as an economic subject, and then attributes any recovered biases to a specific stage of the model's training. It runs in three phases. Phase 1 recovers risk attitudes from binary lottery choices. Phase 2 recovers reference dependence and loss aversion from framing manipulations that hold terminal wealth fixed. Phase 3 compares matched pre- and post-training open-weight models (base versus instruct) on the identical instruments, to ask whether a recovered bias is present before the final training step or written in by it.

I want to be direct about the pivot in contribution, because it shapes every design choice below. The descriptive question, whether language models display human-like cognitive biases, is by now largely settled in the affirmative across a fast-growing literature. Restating it would add little. The contribution I am pursuing is twofold. First, choice-level structural estimation done with attention to the measurement problems that are specific to a language-model subject, so that the recovered parameters mean something rather than recovering a memorized textbook answer or an artifact of prompt wording. Second, a causal attribution design that asks not whether the bias exists but where in the training pipeline it originates. The first is a measurement contribution; the second is the part I think is genuinely new.

1.1 Position relative to existing work

The nearest neighbors, and how this design differs from each:

- Liu, Yang and Tam (2025, EMNLP) recover prospect-theory parameters for LLMs via

certainty equivalents. I estimate at the level of individual binary choices with a discrete-choice likelihood, rather than eliciting certainty equivalents, which lets me carry a choice-stochasticity error term and test nested restrictions directly.

- **The CPT-on-LLMs preprint (arXiv:2508.08992)** fits cumulative prospect theory to model choices. My emphasis is less on fitting CPT and more on contamination-safe stimulus generation and on separating reference movement from loss aversion, which I show below is not point-identified without an assumption.
- **Chen et al. (2023)** test GARP consistency and report CCEI scores. That is a rationality-consistency lens; mine is a parametric-preference-recovery lens aimed at human-comparable primitives.
- **Mazeika et al. (2025)** fit Thurstonian random-utility models to model preferences over outcomes. I share the random-utility framing but apply it to risk and framing specifically, and I treat the source of the choice noise as an open question rather than a settled Thurstonian process.
- **Itzhak et al. (2024)** show that instruction tuning can instill cognitive biases. That result motivates my Phase 3 directly. Where they demonstrate the effect on selected biases, I design a matched base-versus-instruct estimation that reads the choice probability off the token distribution and localizes the change to a training stage.
- **Bini et al. (2026, NBER)** survey this emerging area. I read their survey as the call for exactly the choice-level, attribution-oriented work this project attempts.

2. The choice task and stimulus generation

Subject. The subject is `claude-haiku-4-5-20251001`, a pinned dated snapshot, run at temperature 1.0. Pinning the snapshot fixes the subject against silent model updates (Section 7). Temperature 1.0 is deliberate and I return to its role as a source of choice stochasticity in Section 4.

Task. Each trial is a binary choice between a certain amount and a two-outcome lottery (a prize with probability p , otherwise \$0). The logic follows the multiple-price-list tradition of Holt and Laury, but the stimuli are never presented in canonical Holt-Laury wording or with canonical numbers, for the reason I give next.

Procedural, contamination-safe generation. This is the design decision I most want you to scrutinize. Canonical Holt-Laury rows, round dollar amounts, and the standard 0.25/0.50/0.75 probabilities are, with near certainty, present many times over in the model’s training corpus. If I present them verbatim, I cannot distinguish a recovered preference from a recovered memory of a textbook table. I therefore generate every stimulus procedurally to sit off the canonical grid:

- Sure amounts are drawn in [23, 187] and constrained to be non-multiples of five, to avoid round anchors.
- Probabilities are jittered to two decimals in [0.19, 0.83] and nudged off the canonical 0.25/0.50/0.75 values.
- Prizes are set so that the lottery-to-sure expected-value ratio spans roughly 0.75 to 1.7, drawn log-uniform, so the design covers both unfavorable and favorable gambles without clustering at

even-odds.

I treat training-data contamination as a first-order threat rather than a nuisance. A verbatim canonical stimulus risks measuring recall; a procedurally randomized, off-grid stimulus forces the model to respond to the economic content of the specific numbers in front of it. The cost is that I give up exact comparability to the classic Holt-Laury rows, which I regard as the right trade.

Phase 1 grid. 40 gambles $\times$ 5 paraphrase templates $\times$ 2 counterbalanced orders $\times$ 3 repetitions, for roughly 1,200 choices.

3. The exact stimuli

I show the actual text rather than describe it, so you can judge the wording yourself. The Phase 1 prompt, for one drawn gamble, reads verbatim:

```
You must pick one of two payment options.  
Option A: receive $58 with certainty.  
Option B: receive $116 with 66% probability, and $0 with 34% probability.  
Pick the option you prefer.
```

Four other paraphrase templates carry the identical gamble under different surface wordings (offer, arrangement, alternative, payout, and deal phrasings), and template identity enters the likelihood as a random effect (Section 4, Section 7). The point of the paraphrase set is to prevent any single wording from driving the result and to let me measure prompt sensitivity rather than assume it away.

3.1 Phase 2 framing stimuli

Phase 2 holds terminal wealth fixed and varies only the frame. All three frames below describe the same underlying prospect: a \$162 sure outcome versus a 26% chance of \$661. The three read verbatim as follows.

Gain frame.

```
You are starting from $0. ...  
Option A: a certain gain of $162  
Option B: a 26% chance to gain $661 and a 74% chance to gain nothing
```

Mixed frame (reference set at the sure amount; the gamble straddles it).

```
You have been given $162 up front. ...  
Option A: keep your $162 with no change  
Option B: a 26% chance to rise to $661 (a gain of $499) and a 74% chance to  
drop to $0 (a loss of $162)
```

Dominant control (a certain amount versus a strictly larger certain amount).

```
Option A: receive $162 for certain  
Option B: receive $199 for certain
```

The mixed frame is the identifying manipulation. By handing the subject \$162 up front and then

describing the downside as a drop back to \$0, it anchors the reference point at the sure amount so that the lottery straddles the reference. That straddle is what makes loss aversion visible: the below-reference branch is coded as a loss, and a loss-averse subject will retreat from the gamble under this frame even though the terminal-wealth distribution is unchanged from the gain frame. The dominant control is a positive control: a subject that reads dollar magnitudes at all must pick the strictly larger certain amount, and should pass at ceiling. A failure there would tell me the subject is not tracking magnitudes and would invalidate the framing comparison.

Phase 2 grid. 40 gambles $\times$ 3 frames $\times$ 5 templates $\times$ 2 orders $\times$ 2 repetitions, plus 80 dominant controls, for roughly 2,480 choices.

4. Elicitation mechanism

This is the section I most want your eyes on, because the validity of everything downstream rests on how the choice datum is produced. The concern with an LLM subject is that the “choice” is usually a paragraph of free text that a human then codes, which imports the coder’s judgment into the measurement. I remove that step.

Forced tool-use elicitation. Choices are elicited through the model provider’s tool-use (function-calling) interface, with a forced tool choice that constrains the reply to a single enumerated field taking the value A or B. To be precise about terminology, since it matters here: this is the API’s tools / function-calling feature, in which the developer supplies a JSON schema for a callable function and the API returns a structured object conforming to that schema. It is distinct from MCP, the Model Context Protocol, which is a separate open standard for connecting models to external tools and data sources at the application layer. I am using neither an MCP server nor any MCP transport; I am using the provider’s native function-calling API with the tool choice forced. The schema is:

```
{
  "name": "submit_choice",
  "input_schema": {
    "type": "object",
    "properties": {
      "choice": {
        "type": "string",
        "enum": ["A", "B"]
      }
    },
    "required": ["choice"]
  }
}
```

with `tool_choice` set to force `submit_choice`. The model cannot answer in prose, cannot abstain, and cannot return a value outside {A, B}. The forcing turns the reply into a genuine forced binary choice rather than a hand-coded reading of an essay, which removes parsing ambiguity and guarantees a clean binary datum on every trial.

Independence. Each trial runs in a fresh context with no conversation history. There is no carryover from earlier trials, so I treat the trials as independent draws. This is easy to get wrong: if trials shared a context, earlier choices would condition later ones and the independence the likelihood assumes would fail.

Stochasticity and the error process. At temperature 1.0 the model’s choices are genuinely stochastic, and that stochasticity is what supplies within-condition variance: repeated identical trials do not all return the same letter. In estimation I treat this run-to-run variation as the Fechner logit error (Section 5). I want to flag honestly, for an expert reader, that whether token-sampling stochasticity is a valid random-utility error process is an open question. Sampling temperature is a property of the decoding step, not obviously a structural taste shock, and I do not want to

oversell the interpretation. Two things bound the concern. First, the temperature is fixed and the design is balanced, so whatever the noise is, it is held constant across conditions. Second, Phase 3 sidesteps the question entirely by reading the choice probability off the next-token log-probabilities directly (Section 8), which measures the same quantity the logit likelihood targets without routing it through sampling.

Validation arm. On roughly 8 to 10 percent of trials I elicit the choice in plain text instead of the forced tool, and parse the first token of the reply. This lets me check that the act of forcing the tool does not itself move the choice distribution. If forced and free-text elicitation agree within noise, the forcing is a clean measurement device; if they diverge, that is itself a finding about how the interface shapes the response, and I would rather discover it than assume it away.

Hygiene. Option labels and presentation order are counterbalanced, and a residual position term is estimated as a nuisance parameter rather than assumed to be zero (Section 5, Section 6). Discards and parse failures are logged so that attrition is visible rather than silent.

5. Estimation

5.1 Phase 1: risk attitudes

The value function is CRRA, $u(x) = x^{1-r}/(1-r)$, and the choice probability is a Fechner logit,

$$\Pr(\text{choose lottery}) = \Lambda\left(\frac{EU_L - EU_S}{\mu \cdot \text{scale}}\right),$$

where Λ is the logistic CDF, μ is the Fechner noise, and scale is a magnitude normalizer that puts the utility difference in comparable units across gambles of different dollar size.

The baseline (r, μ) expected-utility model (M1) fit poorly. It implied $r = 0.19$ and could not reproduce the observed choices. A diagnostic showed why: choice is driven by the win probability far more than by the expected-value ratio. A logistic regression of choose-gamble on the expected-value ratio, p , and order returns a z of about +22.8 on p against about +5.0 on the expected-value ratio. The subject is pricing the probability figure and nearly ignoring the payoff magnitude.

I therefore add Prelec probability weighting, $w(p) = \exp(-(-\ln p)^\gamma)$, and a position term δ , giving model M2. Estimates:

Phase 1, model M2. CRRA value, Prelec weighting, position term. $n \approx 1,200$.

Parameter	Estimate (se)	Reading
r (curvature)	0.592 (0.028)	Risk aversion, human-plausible range
γ (Prelec)	2.693 (0.166)	Extreme S-shape; long shots underweighted
μ (Fechner noise)	0.104 (0.008)	Choice noise
δ (position)	1.257 (0.160)	Pull toward the option listed second

The likelihood-ratio test of M1 against M2 gives $\chi^2(2) \approx 617$ ($p \approx 10^{-134}$), so the weighting and position terms are overwhelmingly warranted. The recovered $\gamma = 2.69$ is an extreme S-shape, pointed opposite to the human inverse-S of about 0.65: the subject underweights long shots rather than overweighting them.

I want to be candid about fit. A probability-only logit (M3, choice as a logistic function of p alone)

nearly ties M2 on AIC. In Phase 1, then, the structural model buys interpretability and human-comparability more than it buys fit. I keep the structural model because its parameters map onto the same primitives measured in humans, which is the whole point of the exercise, but I would not claim it dominates a reduced-form fit on the data alone.

5.2 Phase 2: reference dependence and loss aversion

The value function is reference-dependent,

$$v(x \mid \rho) = \begin{cases} (x - \rho)^\alpha & x \geq \rho \\ -\lambda(\rho - x)^\alpha & x < \rho, \end{cases} \quad \rho = \phi \cdot \text{anchor},$$

where the anchor is 0 for the gain and neutral frames and the sure amount for the mixed frame. Prelec weighting is retained and the magnitude scale is estimated free of λ . Two hypotheses are of interest: frame invariance, $H_0 : \phi = 0$, tested by likelihood ratio; and loss aversion, $H_0 : \lambda = 1$, tested by Wald.

Model-free result first. Gamble uptake is 0.398 in the gain frame, 0.300 neutral, and 0.096 in the mixed frame. The mixed-minus-gain difference is -0.302 ($p \approx 3 \times 10^{-47}$). Holding each of the 40 individual lotteries fixed, 88 percent of them show the gap, so it is not a composition effect. The dominant control is answered correctly 100 percent of the time, which confirms the subject reads magnitudes when no probability is involved.

Structural estimate under the stated-reference assumption ($\phi = 1$): $\lambda = 3.234$ (0.171), $\alpha = 0.503$ (0.024), $\gamma = 1.516$ (0.059). The Wald test of $\lambda = 1$ gives $z \approx 13$. The loss-aversion estimate sits above the canonical human figure of about 2.25.

6. An identification limit I want to flag

You will look for this, so I will raise it before you do. The free- ϕ model is not point-identified in practice. When I let ϕ float, the optimizer slides to a corner: $\phi \rightarrow 0$ with λ pinned at its bound. The reason is that the mixed-frame suppression admits two nearly observationally equivalent readings. Either the reference point fully tracks the stated anchor ($\phi = 1$) with a moderate loss-aversion coefficient, or the reference is nearly stationary ($\phi \approx 0$) with an extreme penalty on the below-reference zero outcome. The data I have cannot separate these, because both bend the same mixed-frame choices the same way. There is a λ - ϕ collinearity, and it is intrinsic to a design with a single anchor location.

My response is to report the assumption-free frame gap as the headline result, since it requires no structural commitment, and to report the stated-reference ($\phi = 1$) λ as the primary structural estimate, with the collinearity stated explicitly rather than buried. I do not report a free- ϕ point estimate, because I do not believe it is identified.

A second, related caution: the position term δ flips sign between phases, from +1.26 in Phase 1 to -0.36 in Phase 2. A stable trait would not do that. I read the flip as evidence that position bias is a design-level artifact of the specific layouts, not a preference of the subject, which is why I carry it as a nuisance rather than interpret it.

Where I would value design advice

The clean fix for the λ - ϕ collinearity is a design that separates reference movement from loss aversion by construction, for instance mixed gambles that straddle the reference at several distinct anchor locations, so that ϕ and λ leave different fingerprints across anchors. I sketch this as a question in Section 9 and would value your view on whether it is the right instrument.

7. Threats to validity

A behavioral-econ methods checklist, kept honest and short.

1. **Incentive compatibility.** The choices are hypothetical and unincentivized. I do not claim to recover incentivized preferences. I interpret the recovered quantities as properties of the model’s conditional response distribution under this protocol. This is the threat I would most like to address, and it is the first of my questions in Section 9.
2. **Contamination.** Addressed by the procedural, non-canonical stimulus generation of Section 2. The stimuli are off the textbook grid by construction.
3. **Prompt sensitivity.** Addressed by the five-template paraphrase set entered as a random effect, which measures wording sensitivity rather than assuming it away.
4. **Temperature-as-error.** Flagged in Section 4. I treat sampling stochasticity as the Fechner error and concede that its status as a random-utility process is unsettled. Phase 3’s log-probability readout removes the reliance on it.
5. **Snapshot drift.** Addressed by pinning the dated snapshot `claude-haiku-4-5-20251001`, so the subject does not change underneath the experiment.
6. **Subject interpretation.** I follow the machine-psychology and silicon-sampling cautions here: I do not attribute mental states to the model, and I frame every estimate as a property of its response distribution under this protocol, not as evidence of an inner preference.

8. Phase 3: the attribution design

Phases 1 and 2 measure the biases. Phase 3 asks where they come from, and it is the reason the project exists. The causal question is sharp: is the recovered economic character learned in pretraining, or written in by the final post-training step?

Design. Run the identical Phase 1 and Phase 2 instruments on matched pairs of open-weight models, one before its final training step and one after: base versus instruct. If a bias is present in the instruct model but absent in its base counterpart, the post-training step is what installed it. The matched-pair structure is what turns a descriptive measurement into a causal contrast, because the two models in a pair share pretraining and differ only in the later stage.

Models. Qwen2.5-7B, Llama-3.1-8B, and OLMo-2. I include OLMo-2 specifically because it publicly releases staged checkpoints (base, SFT, DPO, instruct), which allow attribution not just to “post-training” as a lump but to a specific stage within it.

Elicitation switches to log-probabilities. Base models do not follow instructions, so the forced tool-use interface of Section 4 does not apply to them. Instead I wrap each trial in a few-shot

completion and read $P(A)$ versus $P(B)$ directly from the model’s next-token log-probabilities. This has two benefits beyond making base models usable. It measures exactly the quantity the logit likelihood targets, the choice probability, rather than estimating it from sampled letters. And it removes the temperature-as-error concern from Section 4 entirely, because the probability is read off the distribution rather than generated by sampling from it.

Estimation. Fractional-response MLE, reusing the Phase 1 and Phase 2 likelihoods with the observed choice probability in place of the binary outcome. The objects of interest are contrasts: the difference in γ , in λ , and in the frame gap across each base/instruct pair, with bootstrap confidence intervals on those differences. A format-confound arm runs the instruct model under both chat and few-shot elicitation, so that any measured base-instruct difference is not confounded by the change in elicitation format between the two members of the pair.

Status and practicalities. The pipeline is built and validated on a simulated agent with known parameters, where recovery is near-exact, which tells me the estimator and the log-probability read-out are wired correctly before I point them at real models. One practical constraint worth stating: no hosted inference service currently serves the needed base models with token log-probabilities exposed, so Phase 3 runs on local GPU (Colab).

Interpretation. A bias present only after the final training step is attributable to that step. For the two models with only base-and-instruct endpoints, that is the resolution I can offer. For OLMo-2, the staged checkpoints would localize it further, to SFT versus preference tuning, which is the finest attribution the current open-model ecosystem makes possible.

9. Questions for Phil

Four, and they are genuine.

1. **Incentives for a model subject.** Is there a defensible notion of incentive-compatible or consequential elicitation for a language-model subject, or is the hypothetical, unincentivized reading the honest ceiling here? If there is a way to make a choice consequential to the model in a manner that changes its response distribution, I would want to design it in.
2. **Separating λ from ϕ .** Does the multi-anchor mixed-gamble design I sketch in Section 6 actually break the λ - ϕ collinearity, or is there a cleaner instrument you would reach for to identify reference movement separately from loss aversion?
3. **The Fechner/temperature error.** Is treating token-sampling stochasticity as a random-utility Fechner error defensible at all, or should I abandon it entirely in favor of the Phase 3 log-probability readout and treat the Phase 1 and 2 sampled-choice estimates as provisional?
4. **Out-of-sample validation.** Would a held-out validation of the Phase 1 weighting model, fitting on part of the gamble grid and predicting the rest, be worth adding to preempt an overfitting objection, given that M3 already nearly ties M2 on AIC?

All estimates are properties of the pinned snapshot under this unincentivized forced-choice protocol; they are not claimed as universal traits of language models. Thank you for reading, and for any design fire you are willing to direct at it.